# Alex Lee

Waterloo, Canada | a55lee@uwaterloo.ca | LinkedIn | GitHub | Portfolio

## Education

**University of Waterloo** Sep. 2023 - Apr. 2028
*Bachelor's in Data Science* *Waterloo, ON*

**Minor**: Combinatorics and Optimization
**Relevant Courses**: Data Structures, Algorithms, OOP, Statistics, Linear Models

## Skills

**Languages:** Python, SQL(MySQL, PostgreSQL, MongoDB), R, Java, C/C++, Bash
**Libraries:** Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, OpenCV
**Tools:** Git, Docker, Google Cloud, AWS, Linux, Power BI

## Experience

**Machine Learning Engineer | Showcase** Jan. 2026 - Apr. 2026
*City of Edmonton - Future Cities Institute* *Edmonton, AB*

- Developed CityVision, a traffic monitoring project funded by Mitacs, aiming to assisting Edmonton's traffic management and saving **$20,000** for the city.
- Implemented traffic monitoring system using **OpenCV** with **RF-DETR** model to count vehicles and calculate their speed for **20+** case studies, utilizing **Google Cloud Storage** for centralized data management.
- Optimized computational resource efficiency by **33%** by implementing background subtraction module using pixel difference method, to omit parked cars from the study.
- Enhanced speed measurement accuracy by **72%** by calibrating intrinsic and extrinsic parameters to perform 3D reconstruction, transforming perspectives into Bird's Eye View.
- Containerized the entire pipeline using **Docker** and deployed to **GCP Compute Engine**, ensuring environment consistency across development and production stages.

**Database Developer** May. 2025 - Aug. 2025
*Association of Korean Canadian Scientists and Engineers* *Waterloo, ON*

- Led the **team of 6** to develop the official AKCSE website using **Express** and **MongoDB**, improving accessibility to career and academic help resources for students.
- Optimized query efficiency and database scalability by **42%** through implementation of **dynamic CRUD**, while implementing **authentication** and access control features to strengthen data security.

## Projects

**Computer Vision Football Analytics** | *Python, OpenCV, Pandas* | GitHub | Link

- Developed a football analysis system using **YOLOv11** and fine-tuned with **10,000+** images to detect players, referees, and the ball, achieving **0.877** mAP50 with 35.7 ms per-frame latency.
- Implemented **K-means clustering** to automatically assign players to their team, according to their jersey colors.
- Enhanced players' metrics accuracy by **27%** by engineering **camera movement compensation** feature using **Optical Flow** to track metrics consistently even on camera moving environment.
- Deployed services on **AWS EC2** with **SQS**, and built a **Next.js** web app for video upload and real-time analysis.

**Credit Card Fraud Detection** | GitHub

- Developed credit card fraud classification model using **sklearn** and **TensorFlow** with **96%** prediction accuracy.
- Processed and cleaned **200,000+** transaction records, focusing on handling imbalanced financial datasets.
- Utilized **Shallow Neural Network** as a prediction model, comparing **5** different models for optimal results.

**Fullstack Academic Management System** | *Spring, Javascript, HTML/CSS, MySQL, jQuery, Bootstrap* | GitHub

- Built a full-stack academic management web app of **20+** pages with **50+** users using **Spring** and **MySQL**.
- Distributed project on **AWS EC2** with **Apache Tomcat**, with optimized database schema and RESTful APIs.
- Implemented **10+** essential features, **CRUD** operations of course postings and assignments, self implemented real-time chat, custom course search filters, and dynamic operations and views with the database.
- Reduced query response time by **30%** using nested SELECT statements to handle grouped ID increments.
- Ensured reliability with **200+** test cases and resolved issues such as multiple PK, time zone mismatches.